

Module 2

Management Accounting

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)**Answer 1(a)**Revised performance report for last year

	<u>Flexible budget for 150,000 units</u>	<u>Actual costs incurred for 150,000 units</u>	<u>Variance</u>	
	HK\$'000	HK\$'000	HK\$'000	
Sales	<u>16,500</u>	<u>15,840</u>	<u>(660)</u>	A
Cost of goods sold:				
Direct materials	3,000	2,862	138	F
Direct labour	2,250	2,340	(90)	A
Production overheads - variable	1,500	1,500	-	-
Selling overheads - variable	<u>1,200</u>	<u>1,114</u>	<u>86</u>	F
	<u>7,950</u>	<u>7,816</u>	<u>134</u>	F
Contribution margin	<u>8,550</u>	<u>8,024</u>	<u>(526)</u>	A
Production overheads - fixed	3,000	2,800	200	F
Selling overheads - fixed	<u>2,000</u>	<u>3,100</u>	<u>(1,100)</u>	A
	<u>5,000</u>	<u>5,900</u>	<u>(900)</u>	A
Net profit	<u>3,550</u>	<u>2,124</u>	<u>(1,426)</u>	A

Answer 1(b)

Material price variance: $(SP - AP) \times AQ$
 $= [(HK\$3,000,000 \div 150,000 \div 2) - HK\$9] \times (HK\$2,862,000 \div HK\$9)$
 $= HK\$318,000 \text{ F}$

Material quantity variance: $(SQ - AQ) \times SP$
 $= [(150,000 \times 2) - (HK\$2,862,000 \div HK\$9)] \times (HK\$3,000,000 \div 150,000 \div 2)$
 $= HK\$180,000 \text{ A}$

Labour rate variance: $(SR - AR) \times AH$
 $= [HK\$30 - (HK\$2,340,000 \div 80,000)] \times 80,000$
 $= HK\$60,000 \text{ F}$

Labour efficiency variance: $(SH - AH) \times SR$
 $= [(HK\$2,250,000 \div HK\$30) - 80,000] \times HK\$30$
 $= HK\$150,000 \text{ A}$

Answer 1(c)

There was a drop in the unit selling price of HK\$4.40, from HK\$110 ($\text{HK\$}16,500,000 \div 150,000$) to HK\$105.60 ($\text{HK\$}15,840,000 \div 150,000$), possibly due to a deliberate price reduction coupled with an aggressive marketing campaign to boost sales or a deficient sales team failing to maintain the budgeted selling price under fierce competition.

The favourable material price variance may be due to the use of sub-standard materials at a lower price. This may also explain why there was adverse material quantity variance as reflected by higher material wastage because of the use of inferior materials.

The favourable labour rate variance may be due to the hiring of labour with a lower skill level at a lower rate, resulting in an adverse labour efficiency variance. The use of inferior materials may also be a cause of adverse labour efficiency variance.

Answer 2(a)

Statement of profit or loss for current year

	HK\$'000	HK\$'000
Sales ($\text{HK\$}900 \times 40,000$)		36,000
Direct materials	12,000	
Direct labour	9,200	
Production overheads - variable	2,480	
Selling overheads - variable	<u>1,200</u>	
		<u>24,880</u>
Contribution margin		11,120
Production overheads - fixed	3,600	
Selling overheads - fixed	<u>2,516</u>	
		<u>6,116</u>
Net profit		<u><u>5,004</u></u>

Answer 2(b)

Contribution margin ratio:

$$\text{HK\$}(11,120,000 \div 36,000,000) \times 100\% = 30.89\%$$

Break-even in units:

$$\text{HK\$}6,116,000 \div (\text{HK\$}11,120,000 \div 40,000) = 22,000 \text{ units}$$

Margin of safety ("MOS") in percentage:

$$(40,000 - 22,000) \text{ units} \div 40,000 \text{ units} \times 100\% = 45\%$$

The MOS measures the difference between the actual/ budgeted sales and the break-even sales. It is used as a measure of risk: the higher the amount/ units/ ratio, the safer the

situation, since there is a lower probability of reaching the break-even point. A MOS of 45% means that losses begin if the sales revenues of SPL drop by more than 45%.

Answer 2(c)

Estimated unit variable costs:

$$\text{HK\$}(24,880,000 - 12,000,000 \times 5\%) \div 40,000 = \text{HK\$}607$$

Estimated contribution margin per unit:

$$\text{HK\$}(900 \times 97\% - 607) = \text{HK\$}266$$

Estimated sales volume: $40,000 \text{ units} \div 80\% = 50,000 \text{ units}$

	HK\$'000
Total contribution margin ($\text{HK\$}266 \times 50,000$)	13,300
Fixed overheads ($\text{HK\$}6,116,000 + \text{HK\$}1,500,000$)	<u>7,616</u>
Estimated net profit	<u>5,684</u>

Estimated contribution margin ratio:

$$\text{HK\$}266 \div (\text{HK\$}900 \times 97\%) \times 100\% = 30.47\%$$

Estimated break-even in units:

$$\text{HK\$}7,616,000 \div \text{HK\$}266 = 28,631.58 \text{ or } 28,632 \text{ units}$$

Estimated MOS in percentage:

$$(50,000 - 28,632) \text{ units} \div 50,000 \text{ units} \times 100\% = 42.74\%$$

The contribution margin ratio drops slightly from 30.89% for the current year to 30.47% for next year; thus, profitability in terms of contribution remains almost the same.

The MOS in percentage also drops slightly from 45% for the current year to 42.74% for next year, which means the risk of having a loss is comparatively higher but is still at an acceptable level.

Nevertheless, the proposal from the management of SPL is worth taking since it can increase SPL's net profit from HK\$5,004,000 for the current year to HK\$5,684,000 for next year.

Answer 3(a)(i)

Physical unit basis

Product	Output	Cost
Lea	$20,000 \text{ kg} \times \text{HK\$}2,334,000 \div 50,000 = \text{HK\$}$	933,600
Claire	$\underline{30,000 \text{ kg}} \times \text{HK\$}2,334,000 \div 50,000 =$	<u>HK\$1,400,400</u>
	<u>50,000 kg</u>	<u>HK\$2,334,000</u>

Answer 3(a)(ii)

Sales value at the split-off point basis

Product	Sales value at the split-off point	Cost
Lea	$\text{HK\$}1,000,000 \times \text{HK\$}2,334,000 \div 2,050,000 = \text{HK\$}1,138,537$	
Claire	$\text{HK\$}1,050,000 \times \text{HK\$}2,334,000 \div 2,050,000 = \text{HK\$}1,195,463$	
	<u>$\text{HK\\$}2,050,000$</u>	<u>$\text{HK\\$}2,334,000$</u>
	(W1)	

(W1) Lea: $\text{HK\$}50 \times 20,000 = \text{HK\$}1,000,000$; Claire: $\text{HK\$}35 \times 30,000 = \text{HK\$}1,050,000$

Answer 3(a)(iii)

Net realisable value basis

Product	Net realisable value	Cost
Lea	$\text{HK\$}1,100,000 \times \text{HK\$}2,334,000 \div 2,120,000 = \text{HK\$}1,211,038$	
Claire	$\text{HK\$}1,020,000 \times \text{HK\$}2,334,000 \div 2,120,000 = \text{HK\$}1,122,962$	
	<u>$\text{HK\\$}2,120,000$</u>	<u>$\text{HK\\$}2,334,000$</u>
	(W2)	

(W2) Lea: $\text{HK\$}(70 - 15) \times 20,000 = \text{HK\$}1,100,000$; Claire: $\text{HK\$}(54 - 20) \times 30,000 = \text{HK\$}1,020,000$

Answer 3(b)

Further processing decision

Product	Net realisable value HK\$	Sales value at the split-off point HK\$	Incremental profit/ (loss) from further processing HK\$
Lea	1,100,000	1,000,000	100,000
Claire	<u>1,020,000</u>	<u>1,050,000</u>	(30,000)
	<u>2,120,000</u>	<u>2,050,000</u>	

Based on the above analysis, Lea should be further processed while Claire should be sold at the split-off point without further processing.

Answer 3(c)

Revised further processing decision

Product	Net realisable value HK\$	Sales value at the split-off point HK\$	Incremental profit/ (loss) from further processing HK\$
Lea	990,000	1,000,000	(10,000)
Claire	<u>918,000</u>	<u>1,050,000</u>	(132,000)
	<u>1,908,000</u> (W3)	<u>2,050,000</u>	

(W3) Lea: $\text{HK\$}(70 - 15) \times 20,000 \times 90\% = \text{HK\$}990,000$; Claire: $\text{HK\$}(54 - 20) \times 30,000 \times 90\% = \text{HK\$}918,000$

Based on the above analysis, both products should be sold at the split-off point without further processing.

Answer 4(a)

Economic value added ("EVA") = Net operating profit after tax – (Invested capital × WACC)
= $\text{HK\$}828,000 - (\text{HK\$}4,800,000 - \text{HK\$}1,200,000) \times 12\%$
= $\text{HK\$}396,000$

The positive EVA indicates that the economic return on investment is desirable and it increases the business and shareholder's value.

Answer 4(b)

Key performance indicators ("KPIs") reflect organisational processes that are critical to a company's success in implementing its strategy and fulfilling its mission.

Operational KPIs relate to the short-term and focus on the efficiency with which a company is utilising its resources. Operational KPIs could be financial or non-financial in nature.

Strategic KPIs are long-term, firm-specific measures. Strategic KPIs help companies monitor the progress of achieving the strategic objectives that make up their unique corporate and business strategies.

Features of a good KPI:

- Simple and easy to understand
- Readily measurable
- Easy to monitor
- Linked to strategy

* * * END OF EXAMINATION PAPER * * *